IT 340: BUSINESS INTELLIGENCE SYSTEMS

4.5 Project 1 – Baseball Dataset, Part 4

Enhancing Queries with Intermediate SQL

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Executive Summary

Analysis of the Best Batters Using the Lahman Baseball Database

Introduction

Management asked a simple but important question: who are the best batters? To answer it, this analysis used the Lahman Baseball Database, which records the yearly batting results of players across the history of the game.

Rather than relying on a single traditional statistic such as batting average or home run totals, each batter was measured using On-base Plus Slugging (OPS). OPS adds together two measures: how often a batter reaches base (on-base percentage, or OBP) and how much power the batter hits with (slugging percentage, or SLG). Together they give a fuller picture of a batter's value than either measure alone. To make sure the results reflect sustained, proven performance rather than a short hot streak, only batters with at least 3,000 career at-bats were included.

SQL Query

```
WITH clean AS (
    SELECT
        playerid,
        COALESCE(NULLIF(ab, '')::numeric, 0)  AS ab,
        COALESCE(NULLIF(h, '')::numeric, 0)   AS h,
        COALESCE(NULLIF(x2b, '')::numeric, 0) AS x2b,
        COALESCE(NULLIF(x3b, '')::numeric, 0) AS x3b,
        COALESCE(NULLIF(hr, '')::numeric, 0)  AS hr,
        COALESCE(NULLIF(bb, '')::numeric, 0)  AS bb,
        COALESCE(NULLIF(hbp, '')::numeric, 0) AS hbp,
        COALESCE(NULLIF(sf, '')::numeric, 0)  AS sf
    FROM laman.batting
)
SELECT
    pe.namefirst,
    pe.namelast,
    SUM(c.ab) AS Total_At_Bats,
    SUM(c.h)  AS Total_Hits,
    ROUND(SUM(c.h + c.bb + c.hbp) / NULLIF(SUM(c.ab + c.bb + c.hbp + c.sf), 0), 3) AS OBP,
    ROUND(SUM(c.h + c.x2b + 2 * c.x3b + 3 * c.hr) / NULLIF(SUM(c.ab), 0), 3) AS SLG,
    ROUND(
        SUM(c.h + c.bb + c.hbp) / NULLIF(SUM(c.ab + c.bb + c.hbp + c.sf), 0)
        + SUM(c.h + c.x2b + 2 * c.x3b + 3 * c.hr) / NULLIF(SUM(c.ab), 0),
        3
    ) AS Career_OPS
FROM clean c
JOIN laman.people pe ON c.playerid::text = pe.playerid::text
GROUP BY c.playerid, pe.namefirst, pe.namelast
HAVING SUM(c.ab) >= 3000
ORDER BY Career_OPS DESC
LIMIT 10;
```

How the Query Works

- **Cleaning the data.** The first step (called a Common Table Expression) tidies the batting figures. In the early seasons some values are blank, so blanks are treated as zero. This prevents any player's career totals from being lost.

- **Adding up careers.** Each batter's yearly records are totaled into a single career line, using the SUM function.
- **Calculating OBP and SLG.** On-base percentage is hits, walks, and hit-by-pitches divided by the number of plate appearances that count toward it. Slugging percentage is total bases divided by at-bats, so a home run counts for more than a single. The two are then added together to produce career OPS.
- **Protecting against errors.** The NULLIF function prevents a division by zero if a player has no recorded at-bats.
- **Filtering and ranking.** The HAVING clause keeps only batters with at least 3,000 at-bats, and the results are ranked from highest to lowest OPS, showing the top ten.

Query Results

The table below shows the top ten career batters by OPS (minimum 3,000 at-bats).

Rank	Batter	At-Bats	Hits	OBP	SLG	Career OPS
1	Babe Ruth	8,398	2,873	.474	.690	**1.164**
2	Ted Williams	7,706	2,654	.482	.634	**1.116**
3	Lou Gehrig	8,001	2,721	.447	.632	**1.080**
4	Barry Bonds	9,847	2,935	.444	.607	**1.051**
5	Jimmie Foxx	8,134	2,646	.428	.609	**1.038**
6	Hank Greenberg	5,193	1,628	.412	.605	**1.017**
7	Rogers Hornsby	8,173	2,930	.434	.577	**1.010**
8	Mike Trout	5,094	1,543	.415	.587	**1.002**
9	Manny Ramirez	8,244	2,574	.411	.585	**0.996**
10	Mark McGwire	6,187	1,626	.394	.588	**0.982**

Babe Ruth is the best career batter in the dataset, with an OPS of 1.164, which is 0.048 ahead of second-place Ted Williams (1.116). Ruth's lead comes from power: his .690 slugging percentage is the highest of the ten. Williams, by contrast, has the highest on-base percentage (.482), meaning he reached base in nearly half of his plate appearances. The list covers the whole history of the game, from Ruth, Gehrig, and Hornsby in the early 1900s, to Barry Bonds and Manny Ramirez in the 1990s and 2000s, to Mike Trout today. Every batter on the list has more than 5,000 career at-bats, so the ranking reflects long-term excellence rather than a few strong seasons.

Why This Is a Good Metric

1. **It gives a complete picture of hitting.** OPS combines two essential skills: getting on base and hitting for extra bases. A batter who only hits for a high average, or only hits home runs, will not rank as highly as one who does both well.
2. **It is closely tied to winning.** Baseball research has repeatedly shown that OPS tracks closely with the number of runs a batter helps a team score, and scoring runs is what wins games. This links the metric directly to management's goal.
3. **It is easy to read and compare.** OPS is a widely used, standard statistic, and it is simple to interpret. An OPS above .900 is considered excellent, and above 1.000 is historically great. Non-technical stakeholders can trust and understand the numbers immediately.

Limitations of the Metric

1. **It does not adjust for the era.** OPS does not account for the league conditions in which a batter played. The top ten includes players from the early 1900s to today, and the level of scoring, the quality of pitching, and even the baseball itself have changed over time. An OPS of 1.000 in a low-scoring era is not directly comparable to the same figure in a high-scoring one.
2. **It ignores everything except hitting.** OPS measures only what happens at the plate. It does not capture base running, stolen bases, fielding, or the position a player plays, all of which contribute to a player's overall value.
3. **It weighs on-base and power equally.** Simply adding OBP and SLG treats them as equally important. Modern analysis suggests that reaching base is somewhat more valuable than slugging, so OPS slightly favors power hitters. Career totals can also hide the difference between a player's peak and a long, declining tail.

Additional Data That Would Improve the Metric

Several additional sources would make this analysis more reliable and fairer across generations:

- **Ballpark and league adjustments.** Data on the hitting conditions of each stadium and each season would allow a park- and era-adjusted measure such as OPS+, which compares each batter with the league average of his own time.
- **Advanced hitting measures.** Wins Above Replacement (WAR) and weighted runs created (wRC+) would combine hitting with base running and defense to give a full picture of value.
- **Quality of contact and situation.** Modern tracking data such as exit velocity and launch angle, together with clutch performance and results against strong pitching, would show how well a batter hits in the moments that matter most.

Conclusion

Using career OPS with a minimum of 3,000 at-bats, this analysis identifies Babe Ruth as the best batter in the Lahman Baseball Database, followed by Ted Williams and Lou Gehrig. OPS is a strong, easily understood starting point because it rewards both reaching base and hitting for power. It should not be

treated as the final word, however. Adjusting for era and ballpark, and adding measures of base running, defense, and situational hitting, would allow management to compare players from different generations more fairly and to build a more complete picture of batting value.